

Internet of Things

Lab 2

© Fall 2025 – Eng. Yara Ahmed



ESP 32

The board has two connectors located along each side of the board for GPIO, clock, and power line interfaces. Each connector has 19 pins. As shown in Figure 2.2, the two connectors carry the following signals:

The board has a mini USB connector for connecting it to a PC. The board also receives its power from the USB port. Standard +5 V from the USB port is converted into +3.3 V on the board. In addition, two buttons are provided on the board named EN and BOOT, having the following functions

EN: This is the reset button where pressing this button resets the board

BOOT: This is the download button. The board is normally in operation mode where the button is not pressed. Pressing and holding down this button and at the same time pressing the EN button starts the firmware download mode where firmware can be downloaded to the processor through the USB serial port.

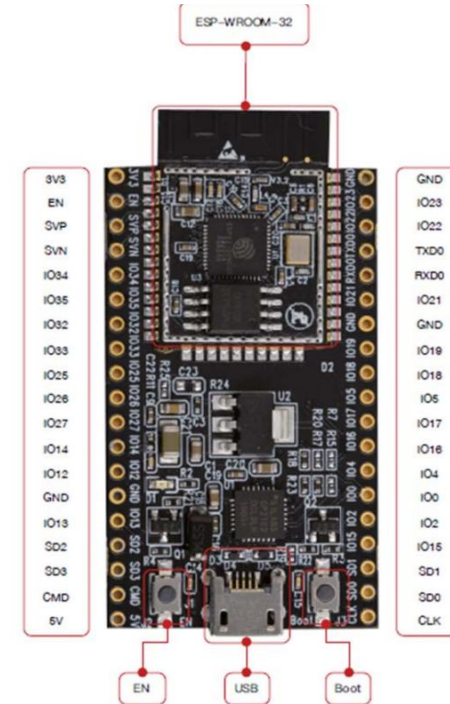
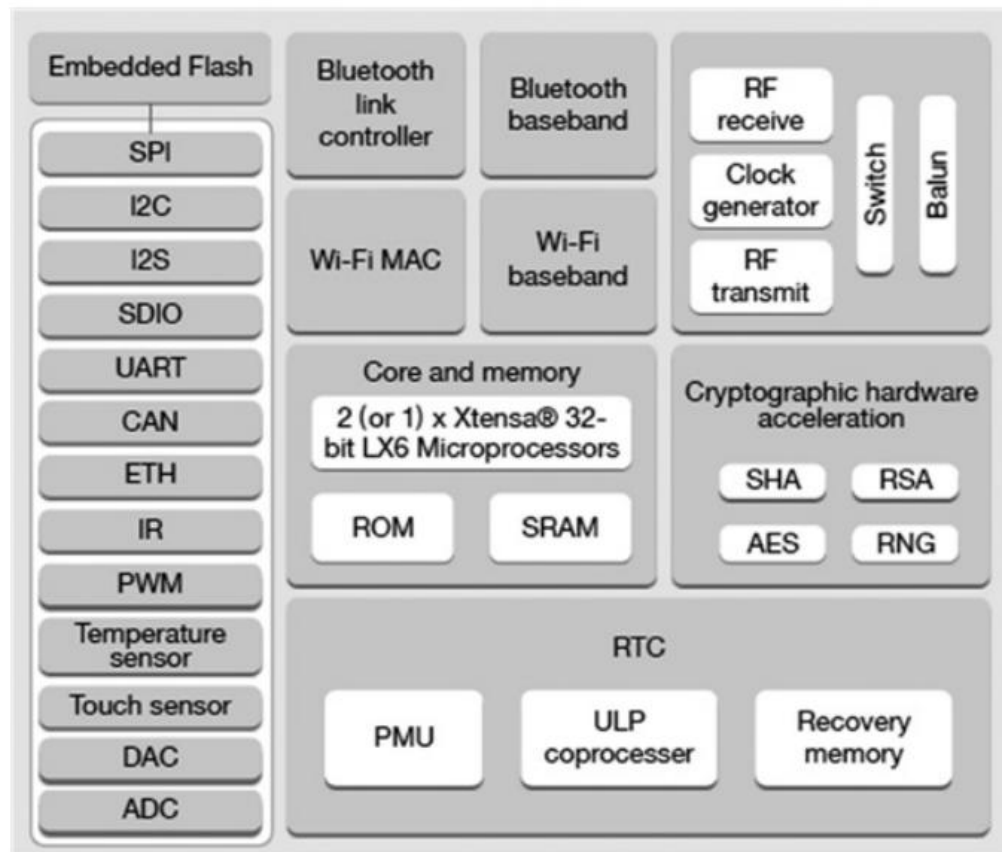


Figure 2.2 ESP32 DevKitC connectors

ESP-32 Development Board WiFi Bluetooth



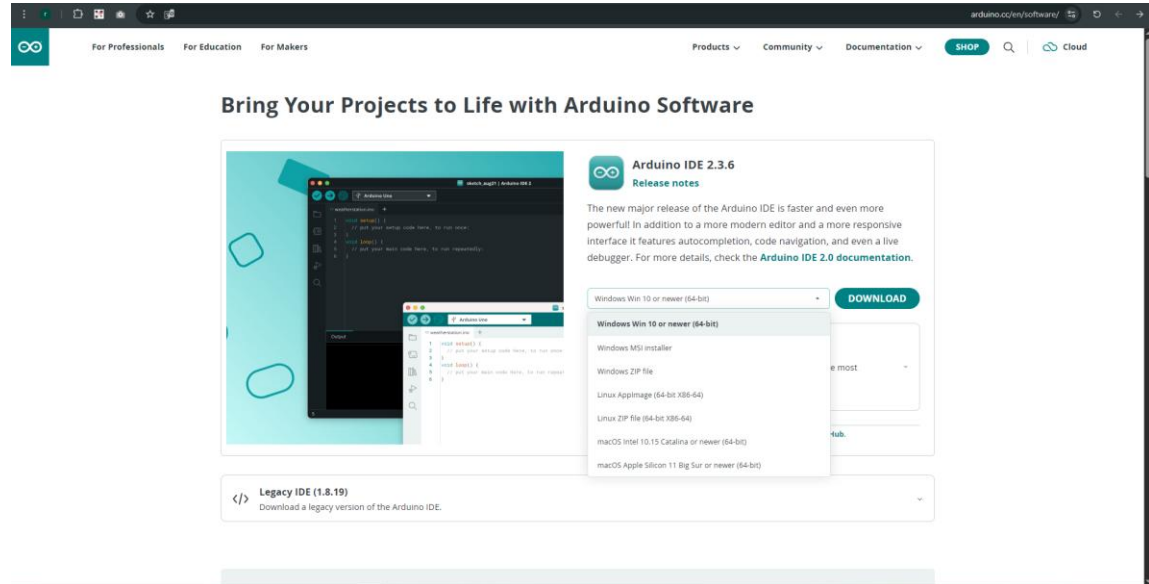
Arduino IDE Download

Step1: Go to this Link <https://www.arduino.cc/en/software/>

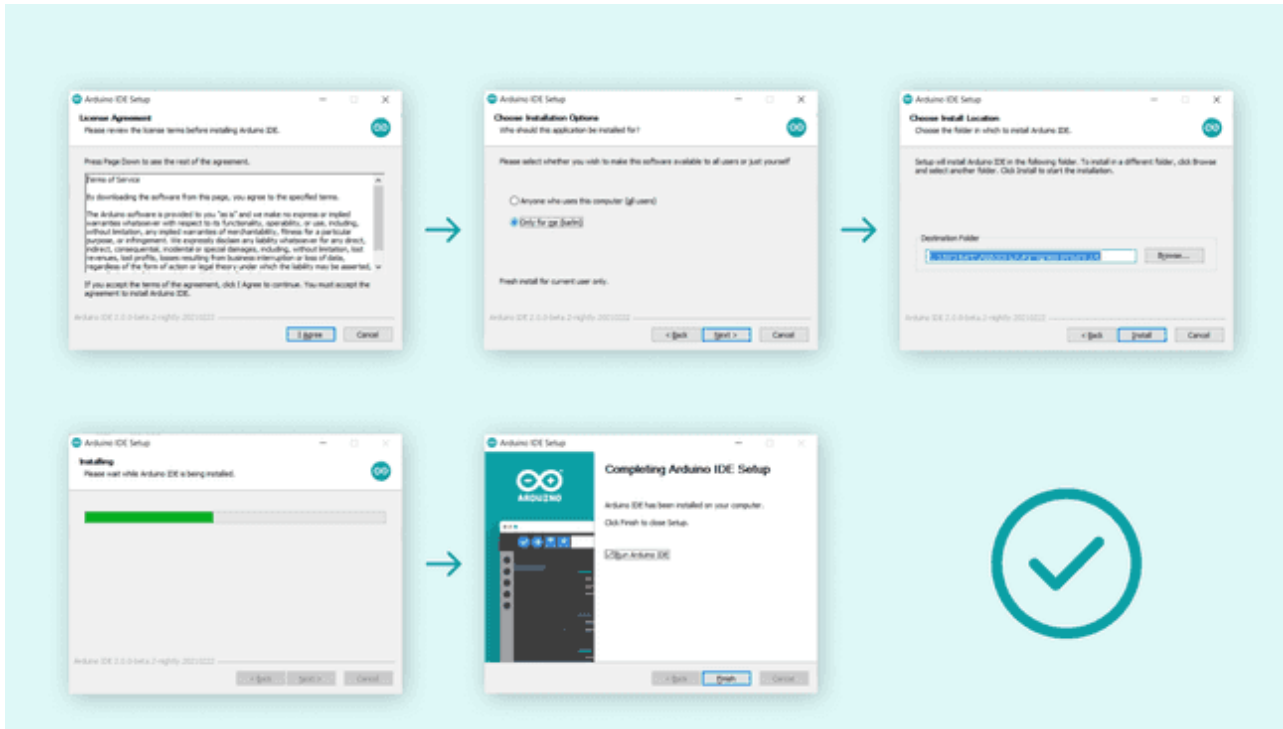
Choose your OS:

- Windows
- macOS
- Linux

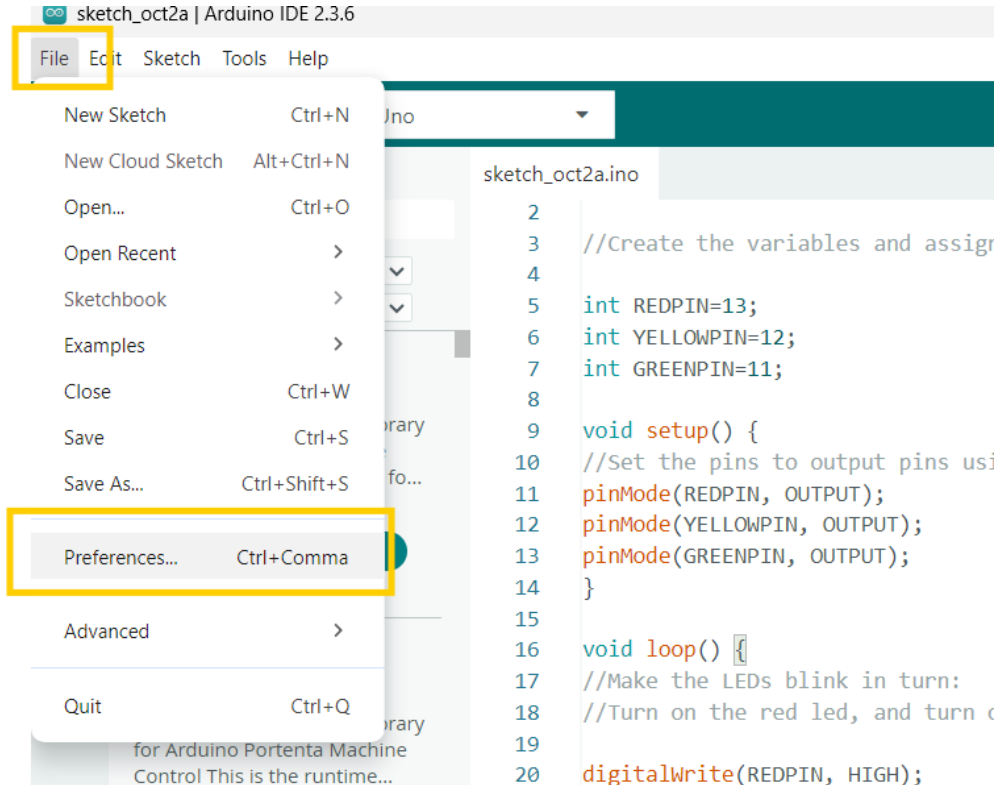
Download the latest version



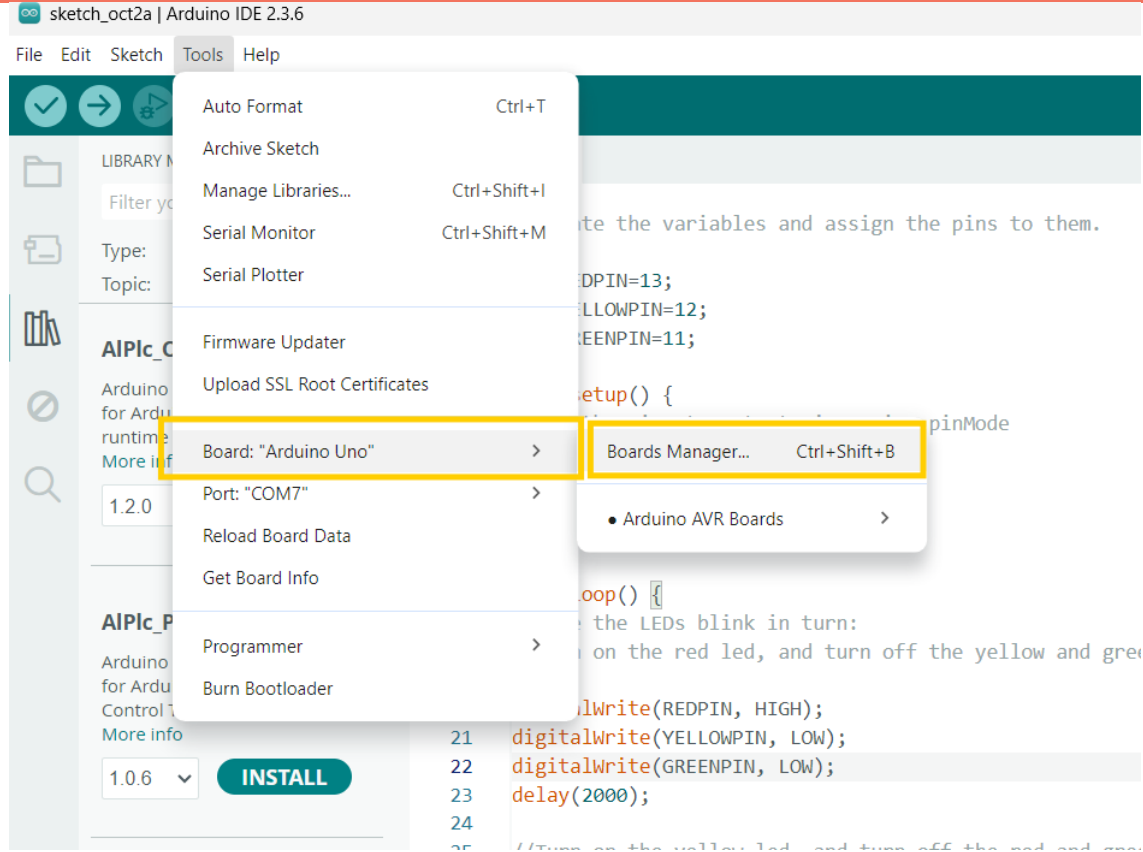
Now we will show steps to setup Arduino IDE.



Arduino IDE



Arduino IDE

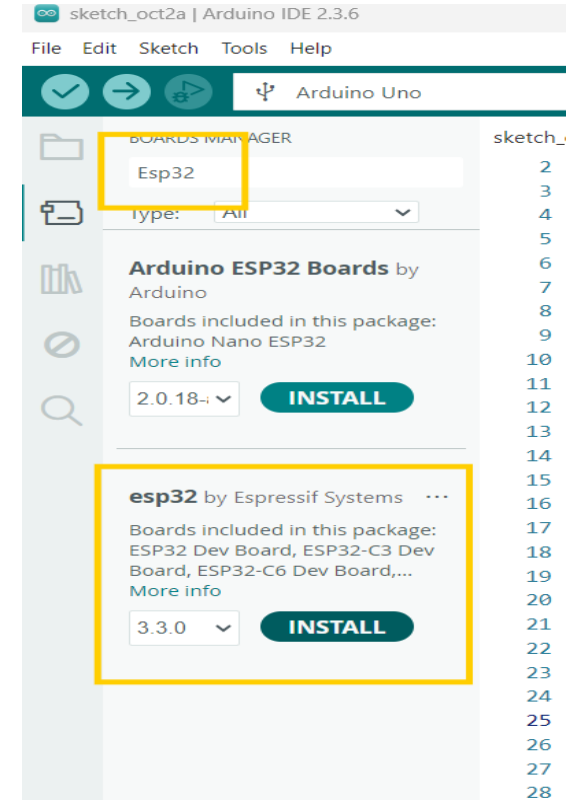


Arduino IDE

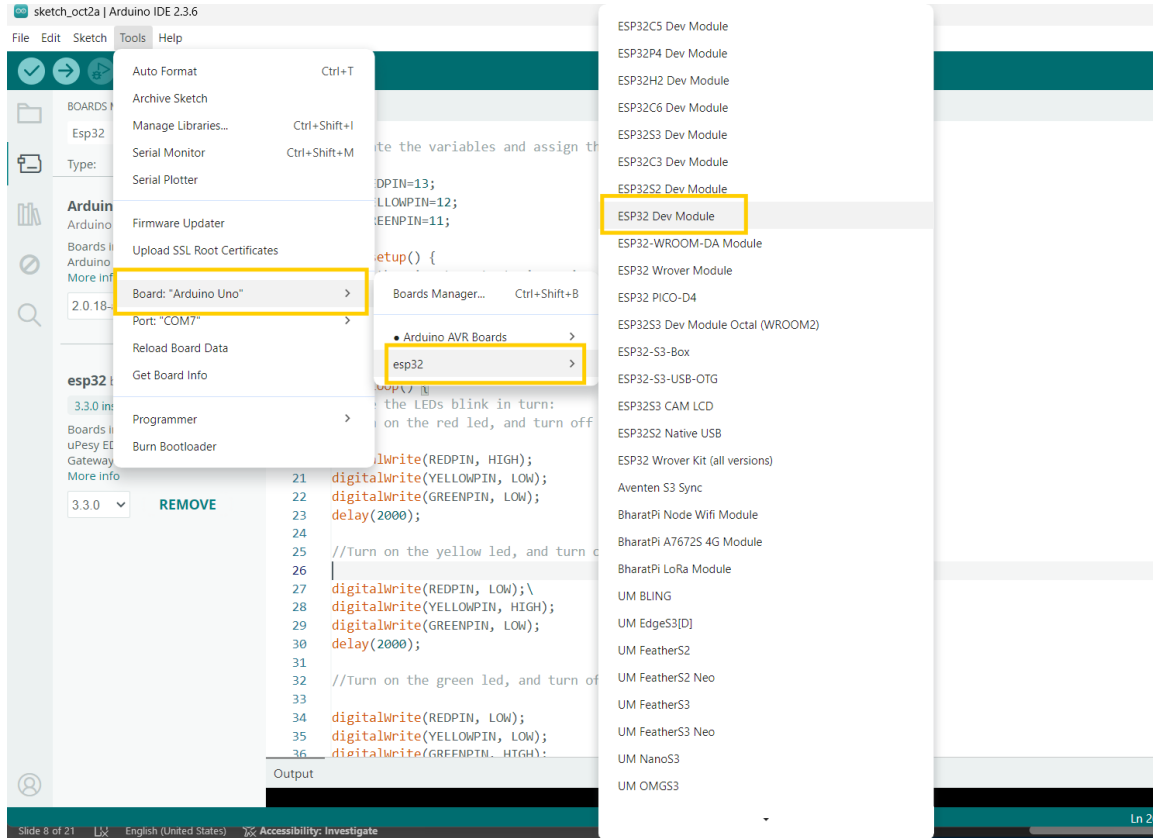
- We should now test the installation to make sure that all the necessary files have been loaded.

We shall make use of one of the supplied example applications to make sure that our program can be uploaded to the ESP32 processor and is working correctly

- Plug in your ESP32 DevKitC to your PC and start the Arduino IDE



Arduino IDE



Project Component we need !

1- ESP-32S 30Pin Development Board WiFi Bluetooth.

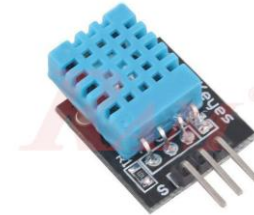


ESP32S - 30PIN

2- breadboard.



3-Temperature & Humidity Sensor Module DHT-11.



4- LEDs.



5- Wires

6- resistor



Getting Started



PROJECT 1 - Flashing LED

Description In this project an LED is connected to GPIO port 23 of the ESP32 DevKitC. The LED is flashed every second.

The Aim The aim of this project is to show how an LED can be connected to a GPIO port and how it can be turned on and off

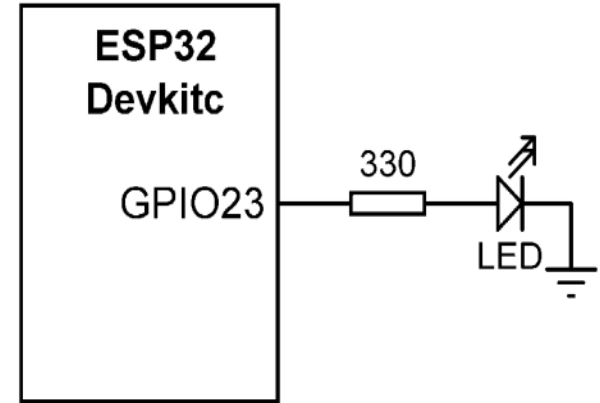


Figure 4.3 Circuit diagram of the project



ESP32 Project Circuit.

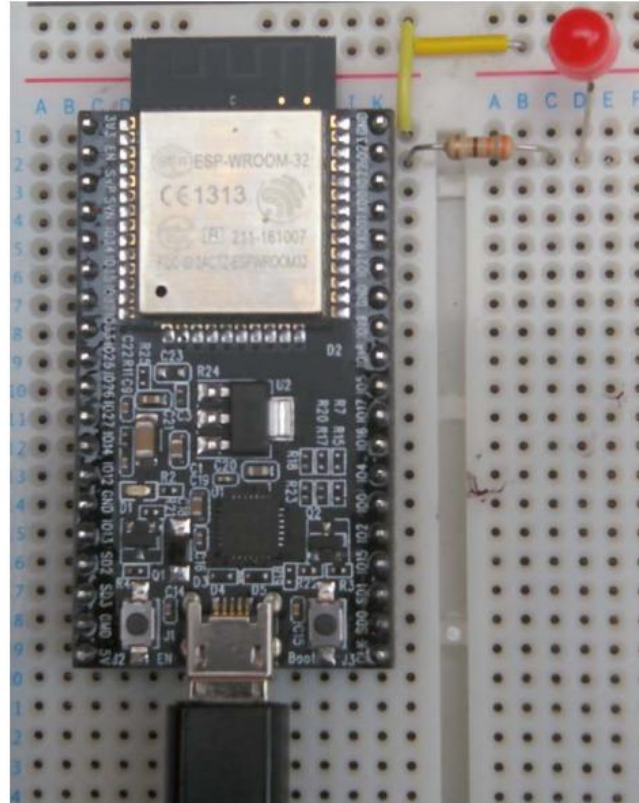


Figure 4.4 Project on a breadboard



PROJECT 1 - Flashing LED

PDL of the Project The PDL of the project is shown in Figure 4.5. The program runs in an endless loop in which the LED is flashed every second.

BEGIN

Assign LED to port pin GPIO23

Configure port pin GPIO23 as output

DO FOREVER

Set LED HIGH

Wait 1 second

Set LED LOW

Wait 1 second

ENDDO

END



Program Listing

The program listing of the project is very simple (program: FlashLED)

Program Description At the beginning of the program variable name LED is assigned to GPIO port 23. Then the LED port is configured as an output port.

The main program is run continuously in a loop, where inside this loop the LED is turned ON (HIGH) and OFF (LOW) with one second delay between each output.

Remember that you should press and hold the BOOT button on the ESP32 DevKitC during the program compile and download process.

Suggestions Modify the program in Figure 4.6 so that the LED ON time is 2 seconds and the OFF time is 5 seconds

```
1  /*****
2  * FLASHING LED
3  * =====
4  *
5  * In this program an LED is connected to port GPIO23 of the
6  * ESP32 DevKitC. The program flashes the LED every second
7  *
8  * Program: FlashLED
9  *
10 *****/
11 #define LED 23
12 void setup()
13 {
14     pinMode(LED, OUTPUT);
15 }
16 void loop()
17 {
18     digitalWrite(LED, HIGH);
19     delay(1000);
20     digitalWrite(LED, LOW);
21     delay(1000);
22 }
23
```

PROJECT 2 – Temperature and Relative Humidity with Serial Monitor

- Description In this project the DHT11 temperature and relative humidity sensor chip is used to get the ambient temperature and the relative humidity. The readings are displayed every second on the PC screen using the Arduino Serial Monitor.
- The Aim The aim of this project is to show how the popular DHT11 relative humidity and temperature sensor chip can be used in projects



PROJECT 2 – Temperature and Relative Humidity with Serial Monitor

Block diagram: Figure 5.9 shows the block diagram of the project.

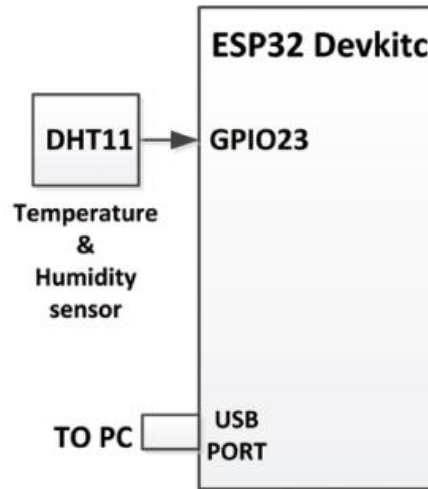


Figure 5.9 Block diagram of the project



PROJECT 2 – Temperature and Relative Humidity with Serial Monitor

- In this project the DHT11 relative humidity and temperature sensor chip is used. The standard DHT11 is a 4-pin digital output device (only 3 pins are used) as shown in Figure 5.10, having pins +V, GND, and Data. The Data pin must be pulled-up to +V through a 10K resistor. The chip uses capacitive humidity sensor and a thermistor to measure the ambient temperature. Data output is available from the chip around every second. The basic features of DHT11 are:
 - 3 to 5 V operation
 - 2.5mA current consumption (during a conversion)
 - Temperature reading in the range 0-50 °C with an accuracy of ± 2 °C
 - Humidity reading in the range 20-80% with 5% accuracy
 - Breadboard compatible with 0.1 inch pin spacings



PROJECT 2 – Temperature and Relative Humidity with Serial Monitor



Figure 5.10 The standard DHT11 chip

In this example the DHT11 module with built-in 10K pull-up resistor, available from Elektor, is used. This is a 3-pin device with the pin layout as shown in Figure 5.11.

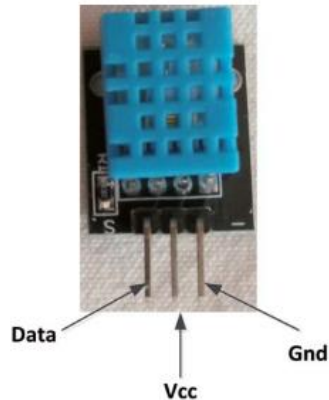


Figure 5.11 Elektor DHT11 module



Construction

- 5 Construction The ESP32 DevKitC board is mounted on a breadboard as shown in Figure 5.13 with the DHT11 sensor chip connected to +3.3 V, GND and GPIO23 of the development board.

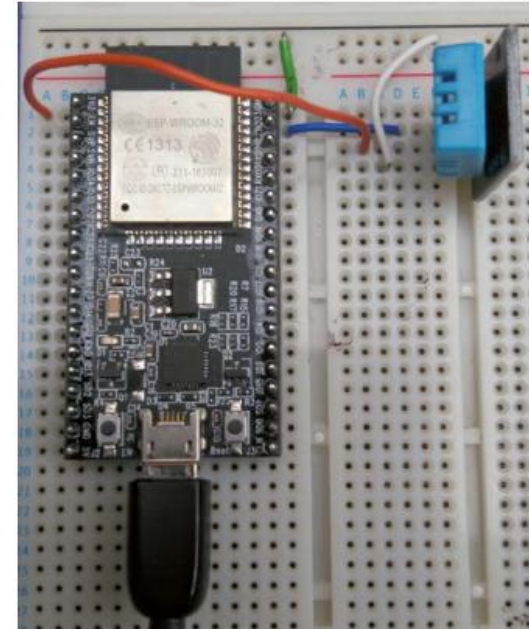


Figure 5.13 Project on a breadboard

PDL of the Project

The PDL of the project is shown in Figure 5.14. The program runs in an endless loop where the temperature and relative humidity are read every second and the readings sent to the Serial Monitor.

BEGIN

```
Include the necessary header files  
Define GPIO23 as the DHT11 port  
Define the DHT type as DHT11  
Set the Serial Monitor baud rate to 9600  
Initialize the DHT library
```

DO FOREVER

```
Read the relative humidity  
Read the ambient temperature  
Display the relative humidity and temperature on Serial Monitor  
Wait 1 second
```

END

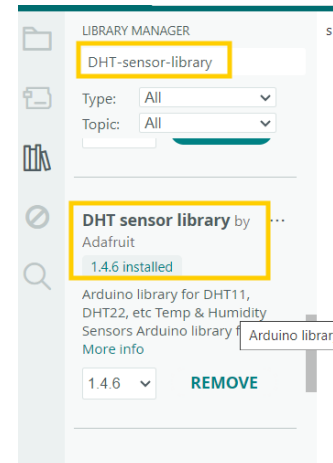
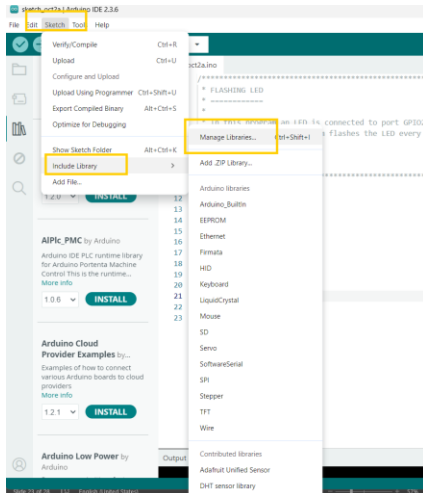
END



Program Listing

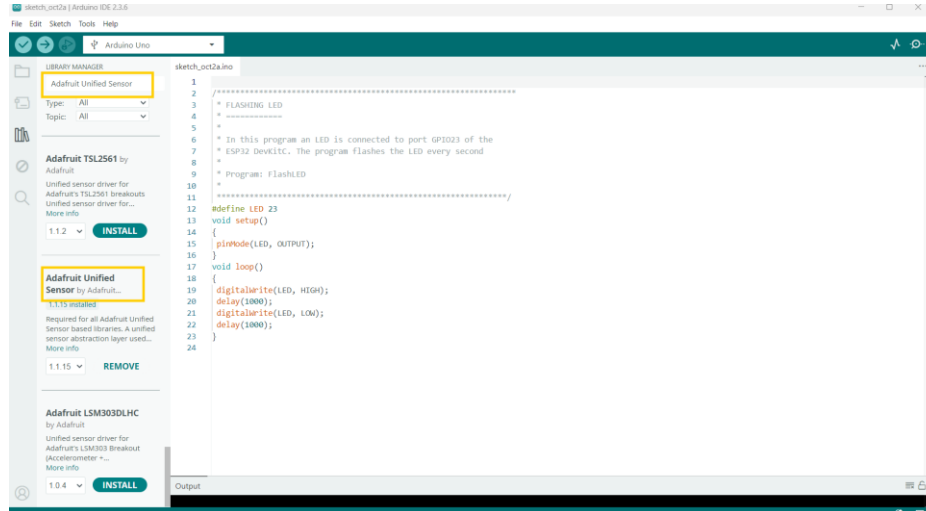
- It is necessary to install the DHT sensor library before the sensor chip can be used in an ESP32 processor based project. The steps to install this library into the Arduino IDE environment are given below

Install the DHT-sensor-library. Click Sketch -> Include Library -> Manage Libraries. • Search for library DHT-sensor-library as shown in Figure 5.15 and click to install the library.



install the Adafruit Unified Sensor library

- We now have to install the Adafruit Unified Sensor library. Click Sketch -> Include Library -> Manage Libraries. • Search for library Adafruit Unified Sensor as shown in Figure 5.16 and click to install the library.



code

```
humidty | Arduino IDE 2.3.6
File Edit Sketch Tools Help
Arduino Uno
humidty.ino
4  *
5  * In this program a DHT11 type relative humidity and temperature
6  * sensor chip is connected to GPIO pin 23. The relative humidity
7  * and the ambient temperature are read every second and the
8  * readings are displayed on the Serial monitor.
9  *
10 * Program: DHT11
11
12 *****
13 #include <Adafruit_Sensor.h>
14 #include "DHT.h"
15 #define DHT11_PIN 23
16 #define DHTTYPE DHT11
17 DHT dht(DHT11_PIN, DHTTYPE);
18 //
19 // Set the Serial Monitor baud rate to 9600 and initialize the
20 // DHT library
21 //
22 void setup()
23 {
24   Serial.begin(9600);
25   dht.begin();
26 }
27 //
28 // Read the relative humidity and the ambient temperature and
29 // display the results on the Serial Monitor
30 //
31 void loop()
32 {
33   float humidity = dht.readHumidity(); // Read humidity
34   float temperature = dht.readTemperature(); // Read temperature
35   Serial.print("Temperature = "); // Display "Temperature="
36   Serial.print(temperature); // Display temperature
37   Serial.print(" Humidity = "); // Display "Humidity="
38   Serial.println(humidity); // Display humidity
39   delay(1000); // Wait 1 second
40 }
41
```



Program Description

- At the beginning of the program header files Adafruit_Sensor .h and DHT .h are included in the program. GPIO pin that the DHT11 is connected to is defined as pin 23, and the DHT type is specified as DHT11.
- Then, the Serial Monitor baud rate is set to 9600 and the DHT library is initialised. The remainder of the program is executed in an endless loop. Inside this loop the relative humidity and the ambient temperature are read and displayed on the Serial Monitor every second. Figure 5.18 shows a sample output displayed on the Serial Monitor



Program Description

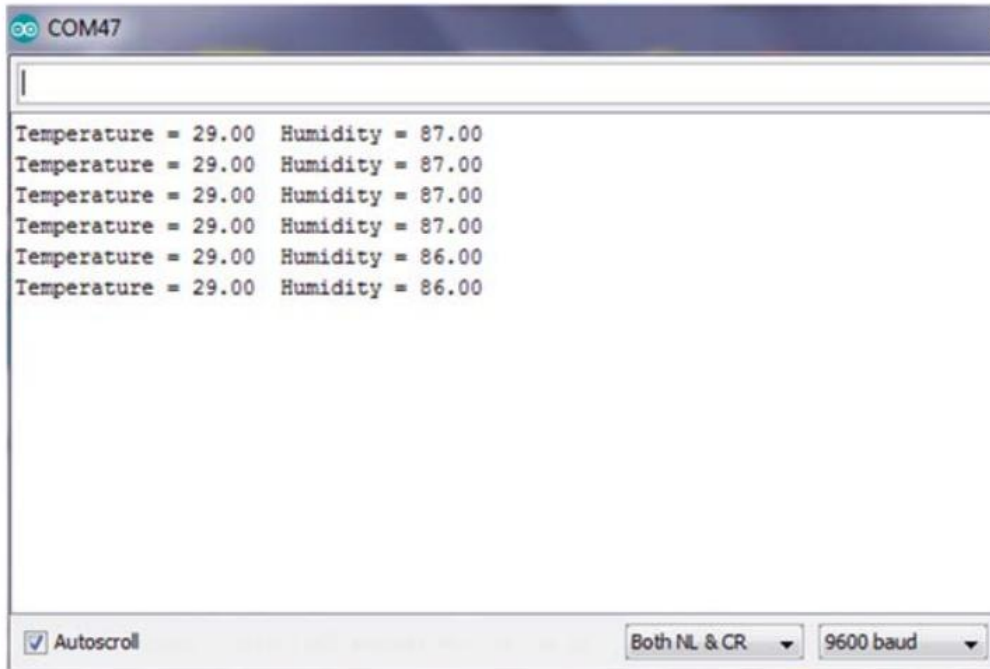


Figure 5.18 Sample output

